

Program: Diploma in Automation and Robotics	
Course Code: 4333	Course Title: Linear IC and Power Electronics
Semester: 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To understand the basics of operational amplifier.
- To comprehend different the linear and non-linear applications of operational amplifiers and Special function IC 555
- To comprehend different type of power electronic components and switches
- To design and analysis power electronic circuits like rectifiers, inverters and DC-DC converters

Course Prerequisites:

Topic	Course Code	Course name	Semester
PN junction diode, rectifier circuit, PNP & NPN transistor, working		Fundamentals of Electrical & Electronics Engineering	

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand Op Amp fundamentals, parameters and configurations.	13	Understanding
CO2	To Understand the linear and non-linear applications of operational amplifiers.	15	Understanding
CO3	Understand the working and characteristics of various power electronic switches	15	Understanding
CO4	Understand the concepts of power electronic converters and to design them based on types of loads and applications	15	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2		3					
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand Op Amp fundamentals, parameters and configurations.		
M1.01	To understand the Internal block schematic of op amp and the ideal characteristics.	1	Understanding
M1.02	To understand the Equivalent circuit of an ideal op-amp and the op-amp parameters.	3	Understanding
M1.03	To understand the open and closed loop configurations.	2	Understanding
M1.04	To understand the working of basic linear circuits.	4	Understanding
Contents: Introduction to operational amplifiers-Internal block schematic of op amp - ideal op-amp - transfer curve - equivalent circuit -Ideal op-amp parameters-CMRR, Slew rate, SVRR, Gain-Bandwidth Product. Open and closed loop configurations-Basic linear circuits-Inverting and Non-Inverting Amplifiers-Virtual ground concept -Voltage Follower.			
CO2	To Understand the linear and non-linear applications of operational amplifiers.		
M2.01	To understand the working of Summing amplifier	1	Understanding
M2.02	To understand the working of comparators and Precision Rectifiers, Differential Amplifiers and Instrumentation Amplifier	3	Understanding
M2.03	To understand the working of Integrator, /Differentiator, Logarithmic and Antilogarithmic amplifier.	2	Understanding
M2.04	To understand the working of Square wave generators (Astable and Monostable Multivibrators) and Sine wave Oscillators (RC phase shift and Wien bridge oscillators)	3	Applying

M2.05	To understand the Functional block diagram of 555 IC	1	Applying
	Series Test – I	1	
Contents: Op amp applications-Summing amplifier- comparator – zero crossing detector- Precision Rectifiers-Differential Amplifiers-Instrumentation Amplifier- Integrator, Differentiator, Logarithmic amplifier and Antilogarithmic amplifier. Square wave generators- A stable and Mono stable Multi vibrators- Oscillators - RC phase shift and Wien bridge oscillators. Specialized IC 555- Functional block diagram.			
CO3	Understand the working and characteristics of various power electronic switches		
M3.01	To understand the operation , characteristics & transistor analogy of a SCR	3	Understanding
M3.02	To explain the role of snubber circuit & latching current and holding current of SCR	2	Applying
M3.03	To explain the working of N-channel & P-channel JFET	2	Understanding
M3.04	To distinguish between MOSFET , JFET & IGBT	2	Applying
M3.05	To compare the characteristics of FET with BJT	1	Remembering
Contents: Power electronics components - characteristics of UJT and its application - SCR construction features -operation – transistor analogy – characteristics - holding current - latching current - turn on time - turn off time – snubber circuits for the protection of SCR – construction & operation of JFET with characteristics , N-channel & P-channel construction-JFET comparison with BJT – MOSFET – comparison of JFET and MOSFET – IGBT – symbols , basic operation			
CO4	Understand the concepts of power electronic converters and to design them based on types of loads and applications		
M4.01	To understand the role of various components like power electronic switches, inductor, capacitor, etc in the operation of converters	1	Understanding
M4.02	To explain the operation of rectifier circuit, half bridge and full bridge rectifier circuit with various loads	3	Understanding
M4.03	To explain the working principle & operation of inverter circuit in half bridge and full bridge mode	3	Applying
M4.04	To understand the design concept and role of various components in DC- DC converters	3	Applying
	Series Test – II	1	
Contents: CONVERTERS - AC – DC rectifiers, half bridge, full bridge rectifiers with R, RL,RLE load &freewheeling diode(Uncontrolled & Fully controlled rectifiers). DC-AC			

inverter (single phase half bridge inverter, waveforms, principles of operation - single phase full bridge inverter , principles of operation, waveforms), DC-DC boost converter and buck converter (design , working principle and waveforms)

Text / Reference:

T/R	Book Title/Author
R1	P.S. Bimbira. Power Electronics; Khanna publishers
R2	Dr. B.R. Gupta V & Singhal. Power electronics; S.K. Kataria & sons
R3	S.K. Bhattacharya. Fundamentals of Power Electronics; Vikas Publication House Pvt Ltd
R4	Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, 3/e, 2010
R5	Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Prentice Hall, 4/e, 2010
R6	Salivahanan S. and V. S. K. Bhaaskaran, Linear Integrated Circuits, Tata McGrawHill, 2008.

Online Resources:

Sl No	Website Link
1	https://nptel.ac.in/courses/108/101/108101038/
2	https://youtube.com/playlist?list=PLp6ek2hDcoND7i5-DAD9mPmYF1Wg6ROdO
3	https://nptel.ac.in/courses/108/101/108101038/
4	https://youtube.com/playlist?list=PLp6ek2hDcoND7i5-DAD9mPmYF1Wg6ROdO